

Answers

1. B	2. 2	3. D	4. 2	5. D	6. 2	7. B	8. 2
9. B	10. C	11. C	12. 40	13. 4			

Explanations

1. **B**

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that among the options station E is visited the largest number of times.

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2.2

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that the teams have passed through B 2 times in this given period.

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3.D

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that at 10.15 hrs, team 3 is travelling from station D to station E.

The correct option is D

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4.2

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that team 4 passed station E 2 times in a day

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5.D

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that 2 teams (teams 3 and 4) have passed through station C on the given day.

The correct option is D

 VIDEO SOLUTION



6.2

Since there are two routes i.e A-M-B and A-N-B and four cars, then 2 cars must be allowed to take each route. In case if one car tried to break rule, then its travel time will increase. Now assume that on route A-M-B three cars are allowed and on route A-N-B one car is allowed, then one car running on A-M-B can break the rule and reduce its travel time. Hence, two cars must be allowed on each route.

7. B

Since, two cars are allowed on each route, The A-M part and N-B will take same time. The difference will be in travelling M-B part and A-N part, and that difference is 0.1 minute. In route M-B car will take $20+0.9 = 20.9$ min and in route A-N car will take $20+1 = 21$ min. So difference = 0.1 min

8. 2

Case 1: Let us assume 1 car takes AMB route, 3 cars take ANB route

Then travel time of AMB will be $A-M + M-B = 6 + 20 = 26$

Then travel time of ANB will be $A-N + N-B = (20+2) + (6+3*2) = 34$

Now, one car(A) travelling on ANB broke the rule and decided to move on AMB route. then

Case 2: Let us assume 2 cars take AMB route, 2 cars take ANB route

Then travel time of AMB will be $A-M + M-B = (6+3) + (20+0.9) = 29.9$

Then travel time of ANB will be $A-N + N-B = (20+1) + (6+3) = 30$

Since the car A reduced its time from 34 to 29.9, the case-1 route is not optimal, hence case 1 is invalid. Now, one car(B) travelling on ANB broke the rule and decided to move on AMB route. then, case

Case 3: Let us assume 3 cars take AMB route, 1 car take ANB route

Then travel time of AMB will be $A-M + M-B = (6+3*2) + (20+0.9*2) = 33.8$

Then travel time of ANB will be $A-N + N-B = (20) + (6) = 29$

Now, one car(C) travelling on AMB broke the rule and decided to move on AMNB route. then

Case 4: Let us assume 2 cars take AMB route, 1 car take AMNB route and other take ANB route.

Then travel time of AMB will be $A-M + M-B = (6+3*2) + (20+0.9) = 32.9$

Then travel time of AMNB will be $A-M + M-N + N-B = (6+3*2) + (7) + (6+3) = 28$

Then travel time of ANB will be $A-N + N-B = (20) + (6+3) = 29$

Since this car(C) reduced its time from 33.8 to 28, the route is not optimal, hence case 3 is invalid.

Since this car(B) reduced its time from 30 to 28, the route is not optimal, hence case 2 is invalid.

Now, one car(D) travelling on AMB broke the rule and decided to move on AMNB route. then

Case 5: Let us assume 1 car takes AMB route, 2 cars take AMNB route and other takes ANB route.

Then, the portion A-M will be travelled by 3 cars, M-B by one car, M-N by 2 cars, A-N by 1 car and N-B by 3 cars.

Then, travel time of AMB will be $A-M + M-B = (6+3*2) + (20) = 32$

Then, travel time of AMNB will be $A-M + M-N + N-B = (6+3*2) + (7+1) + (6+3*2) = 32$

Then, travel time of ANB will be $A-N + N-B = (20) + (6+3*2) = 32$

It is clear that the car D reduced its time from 32.9 min to 32 min if it broke the rule. Hence, case 4 is invalid.

In this arrangement of case 5, no car can improve their travel time by changing their path

Hence, the optimal allocation will be to order 2 cars on A-M-N-B route.

9. B

From the previous question we have found that

1 car take AMB route, 2 cars take AMNB route and other take ANB route.

Then the portion A-M will be travelled by 3 cars, M-B by one car, M-N by 2 cars, A-N by 1 car and N-B by 3 cars.

Then travel time of AMB will be $A-M + M-B = (6+3*2) + (20) = 32$

Then travel time of AMNB will be $A-M + M-N + N-B = (6+3*2) + (7+1) + (6+3*2) = 32$

Then travel time of ANB will be $A-N + N-B = (20) + (6+3*2) = 32$

The minimum travel time from A to B is 32 min.

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10. C

There are ten cities. We need to find the minimum number of flights required to travel from any city to any city. Any two cities can be selected in $10C2$ ways. Now for these two cities, a person will need minimum 4 flights. (1 to go from A to B, 1 to go from B to A. Similarly, 1 to return to A and 1 to return to B) Thus, minimum number of required flights = $45*4 = 180$.

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11. C

From each hub, there will be flights to 7 cities. So total total number of flights originating or terminating at each hub = $7*4 = 28$. For all three hubs, it would be $28*3 = 84$

There are three hubs in total. Each hub must also be interconnected. The total number of flights between any two hubs will be 4. For three hubs it will be 12.

Hence, the required number will be $84 + 12 = 96$.

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12. 40

In G1, we have three cities namely A, B, C. Person living in any of these three cities should be able to travel to other city once in the morning nad one in the night. Therefore, a total of 4 flights are required between a pair of cities.

A $\rightarrow$ B (Morning flight)

A $\rightarrow$ B (Evening flight)

B $\rightarrow$ A (Morning flight)

B $\rightarrow$ A (Evening flight)

Number of flights between the cities of G1 = $3c2*4 = 12$

Between cities in A and any city in G2 = $3*4 = 12$

Between B and any city in G3 = $2*4 = 8$

Between C and any city in G4 = $2*4 = 8$

Total = $12*2 + 8*2 = 40$

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13. 4

The cities those were a part of G2 will be shifted to either G3 or G4 but that will not have any impact on the number of the total flights from G1. The only reduction which will take place due to the number of flights shutting down from A to C.

Hence, the maximum reduction in the number of direct flights as compared to the situation before the operational difficulties arose = 4

Alternate method :

Let us determine the number of flights under new conditions.

Flights between A and B = 4

Between B and any city in G3 = $4 \times 4 = 16$

Between C and B = 4

Between C and any city in G4 = $4 \times 4 = 12$

Hence, total flights = 36

Thus, reduction = $40 - 36 = 4$.

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